

Open PCIe on GateMate

From a finished Root Complex on Artix-7, to a soft PCIe stack that anyone can simulate.

WHERE WE CAME FROM

openPCIe is finished

On Artix-7 the whole protocol stack was already inside the Xilinx hard macro. On GateMate, only the SerDes is.

FINISHED

openPCIe · Artix-7

TL	hard macro
DLL	hard macro
MAC / LTSSM	hard macro
PIPE	hard macro
SerDes	hard macro

IN PROGRESS

openCologne-PCIe · GateMate

TL	to write
DLL	open RTL · needs TL to fully test
MAC / LTSSM	open RTL
PIPE	open RTL
SerDes	hard macro + our model

THE BLOCKER

The SerDes model could not be shared

Built from CologneChip's real chip RTL, so it is their IP. Nobody outside the NDA could run any simulation at all.

133 FILES	29k LINES	25 ps INTERNAL CLOCK STEP	NDA CANNOT PUBLISH
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WHAT WAS BUILT

An open behavioural model

Drop-in replacement. Same ports, same parameters, no vendor code. It describes the behaviour, not the chip.

1 FILE, VS 133	1.8k LINES, VS 29K	0 LINES OF VENDOR CODE	18× FASTER
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SPEED

Same test, same 101 checks

Zero errors with either model.



624 module instances and 11,905 logic nodes → 8 and 225.

CONFIDENCE

Proved, not hoped

Three independent checks. Together they found three real bugs that ordinary testing missed.

Side by side 400 words · 0 differences Both models, same simulation, same data. Delay matches to 0.0 ns.	Cross-linked 0 errors, both directions Our transmitter into their receiver, and back the other way.	Formal proof 6 proved · 0 failed SAT solver over every possible input, not a sample.
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RUNNING TODAY

The full stack, over a real serial link

Packets from the data link layer are encoded, sent one bit at a time, recovered and decoded – then checked. 1.4 seconds, zero errors.

dll_logic	CologneChip · unchanged
MAC	link training
ccfpga_pipe_logic	CologneChip · open RTL
ccfpga_serdes	ours · open behavioural model
serial loopback – 2.5 Gb/s	
1 0 1 1 1 1 1 0 0 1	

Nothing in this stack needs an NDA.

REMAINING WORK

What still has to be added

□ TL – transaction layer	our next piece
□ Software and PIO test application	not started

Measured on the project's own test cases Icarus Verilog 12 · yosys 0.30